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RESEARCH INTERESTS

Mechanistic Interpretability · AI Alignment & Safety · LLM Evaluation · Multi-Agent Safety

EDUCATION

SRM Institute of Science and Technology, Ramapuram *Chennai, India*
B.Tech, Computer Science & Engineering (AI & Machine Learning) CGPA 8.7 / 10.0 · Expected May 2028

RESEARCH PROJECTS

Collusion Monitors in Agent Swarms (Apart Research Swarm Alignment Hackathon, solo) Nov 2026
Python · Ollama · small open-weight LLMs · AI Village agent logs

- Tested five behaviour-based collusion monitors on 227 repeated-pricing games played by small LLMs in swarms of 4, 8 and 12 agents.
- Found monitors accused innocent same-model pairs 4–16× as often as mixed-model pairs; games with a false accusation rose from 60% at 4 agents to 100% at 12.
- Calibration on known-innocent pairs or label-free twin replays cut false accusations to 15–50%; on real AI Village logs, a label-free control removed a timing-only same-model signal (13 of 22 pairs flagged).

Detecting Collusion in LLM Agents (Apart Research AI Collusion Sprint, solo) Oct 2026
Python · open-weight LLMs · statistical calibration

- Tested whether pairwise net influence (Bonjour et al., 2022) detects collusion among LLM agents in repeated three-player rock-paper-scissors; ran 64 games on Qwen3-8B, Llama 3.1 8B and gpt-oss-20b.
- Found it blind when teammates collapse to one move, and found it falsely accused the two same-model players in 4 of 5 mixed-model games; a same-model leave-one-out calibration cut false accusations from 38% to 6%.
- Reported small samples and post hoc labels as limitations. Paper and public code submitted; judging pending.

GPT-2 Small Mechanistic Interpretability Pipeline 2026
Python · PyTorch

- Built a complete GPT-2 Small forward pass from scratch, with hooks capturing activations across all 144 attention head pairs.
- Planted a Previous Token Head (L3H0) and Induction Head (L5H1) with synthetic weights; validated by activation patching (PTH score 0.87) and residual-stream causal interventions.

ConspiracyBench (Stationhouse), open-source contributor Oct 2026
Python · Inspect AI · Hugging Face

- 40 merged PRs: stress-tested the benchmark's Inspect evaluation pipeline (incomplete transcripts, bounded retries and timeouts, malformed-input handling, a Windows encoding fix) and added sourced benchmark questions.
- Found that a small local LLM judge could be flipped by a reply imitating a grader note (exploratory, n=1 per case).

CPU-Only VQA Benchmark for Lightweight Vision-Language Models 2026
Python · PyTorch · HuggingFace Transformers · Matplotlib

- Benchmarked three sub-2B-parameter VLMs (SmolVLM-500M, Moondream2-1.8B, TinyLLaVA-1.4B) on a stratified 1,013-question VQAv2 subset under CPU-only, 8GB-RAM constraints; reported accuracy, latency, peak RAM, load time and throughput.
- Found, with a chi-square test ($p < 0.000001$), a consistent accuracy drop on numeric questions across all three models; validated pairwise comparisons with McNemar's exact test over 3,000+ predictions.
- Built a resumable pipeline with automatic checkpointing for multi-day laptop runs. Manuscript in preparation for a Scopus-indexed venue.

DistilBERT Sentiment Classifier 2026

- Fine-tuned DistilBERT-base on SST-2 (AdamW, cosine LR decay) to 89% test accuracy with a hand-written training loop. Also implemented backpropagation from scratch in NumPy.

TECHNICAL SKILLS

Languages	Python, C
ML / DL	PyTorch, HuggingFace Transformers, scikit-learn
Research	LLM evaluation (Inspect AI), activation patching, causal intervention, statistical testing
Tools	Git, GitHub, Linux, Jupyter, LaTeX, TransformerLens

RELEVANT COURSEWORK

Machine Learning · Deep Learning · Linear Algebra · Probability & Statistics · Data Structures & Algorithms · Natural Language Processing